

Statistical Learning Applications to European Football Data

Course: Statistical Learning and Large Data I e II
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What drove our research?

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We wanted to apply statistical learning to answer three different questions related to European football

Premise: we LOVE football, but we wanted to know more about it

The team analyst) Are there similar playing styles in Europe? Do they change between different national championships?

The talent scout) Are there specific skills to focus on when scouting future football talents?

The gambling addict) Is it possible to beat the betting market? Where are efficiency deviations concentrated?

Research questions and dataset description



Can European football teams be grouped into distinct playing styles using unsupervised learning techniques?

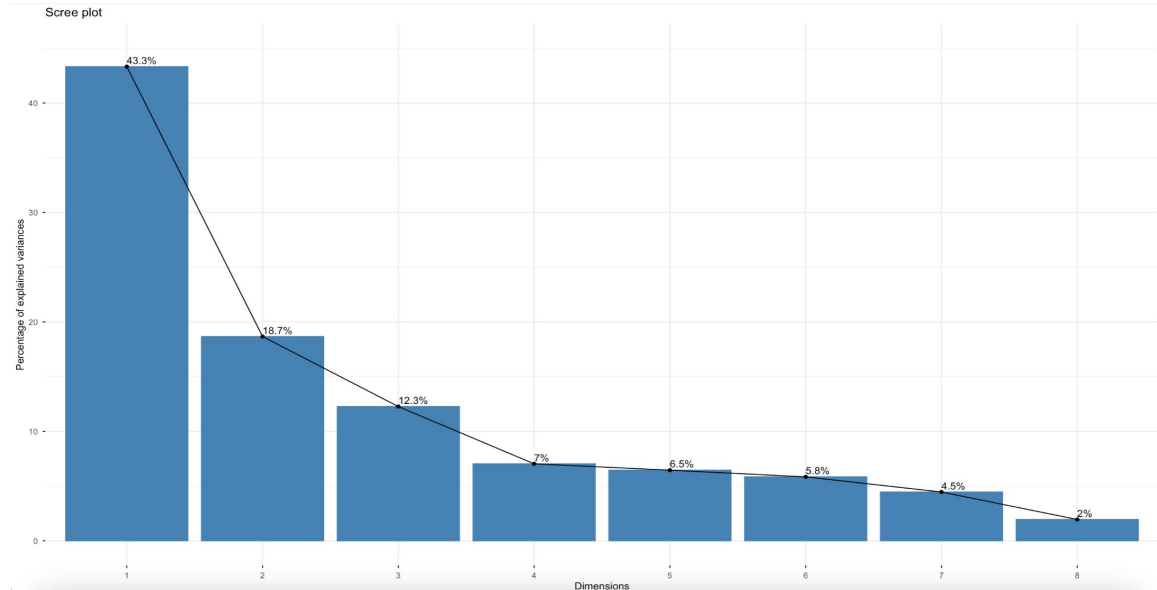
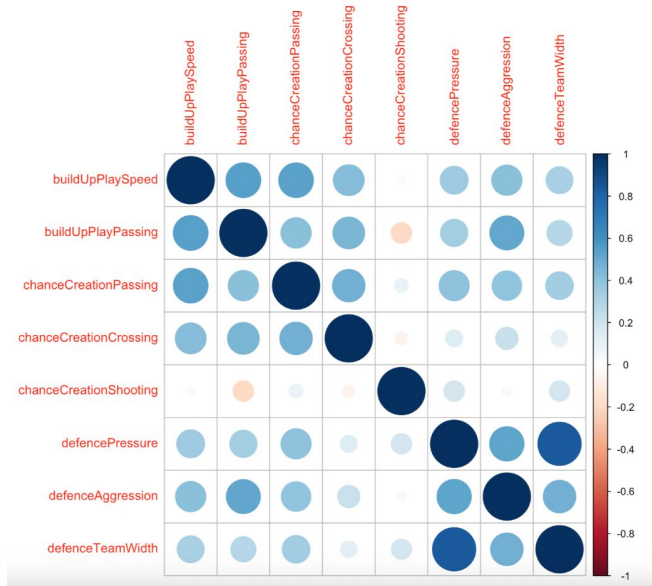
Dataset: European Soccer Database

Content: matches, teams, players and technical/tactical attributes

Focus of the analysis: Team_Attributes

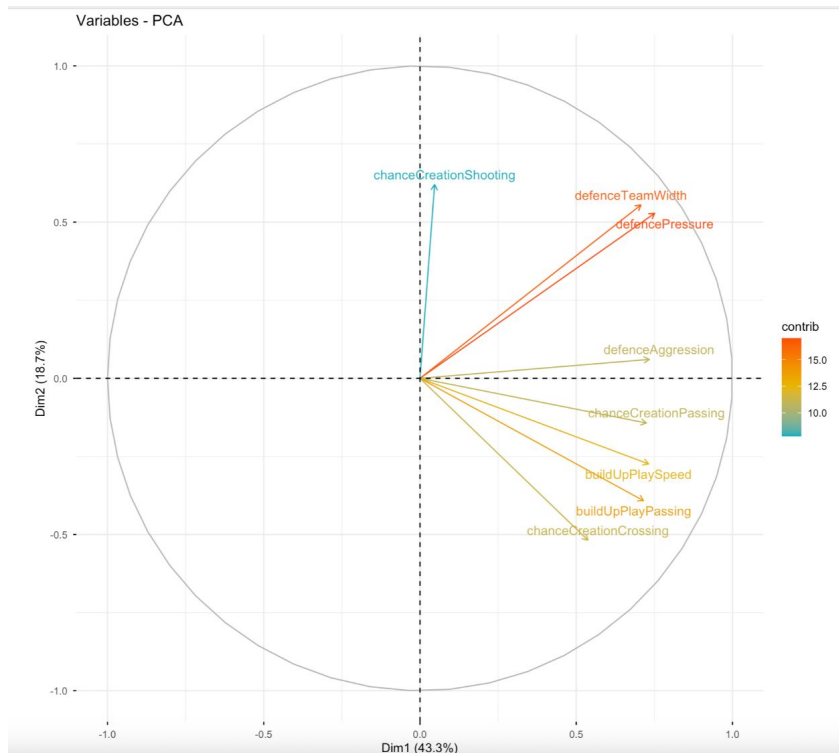
Variables used: build-up, passing, crossing, shooting, defensive pressure, aggression and team width

Correlation matrix and PCA explained variance

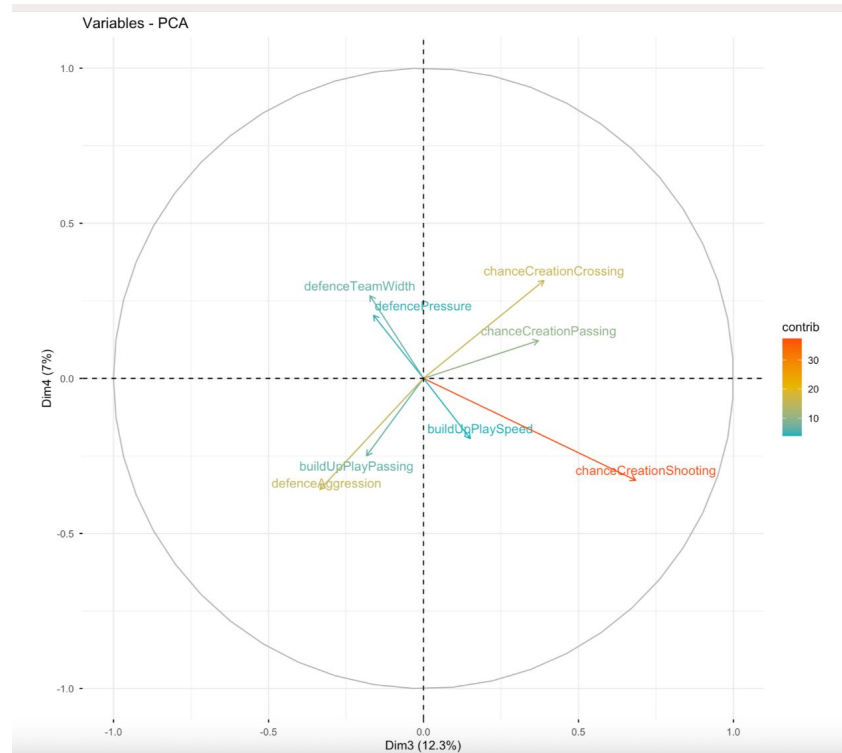


- Tactical variables are not independent: several of them capture related aspects of playing style.
- PCA reduces redundancy by transforming the original variables into uncorrelated components.
- The first three principal components explain approximately 74% of total variance which is close to the conventional threshold of 75%.
- A few dimensions capture most of the tactical information.

Interpretation of PCA components

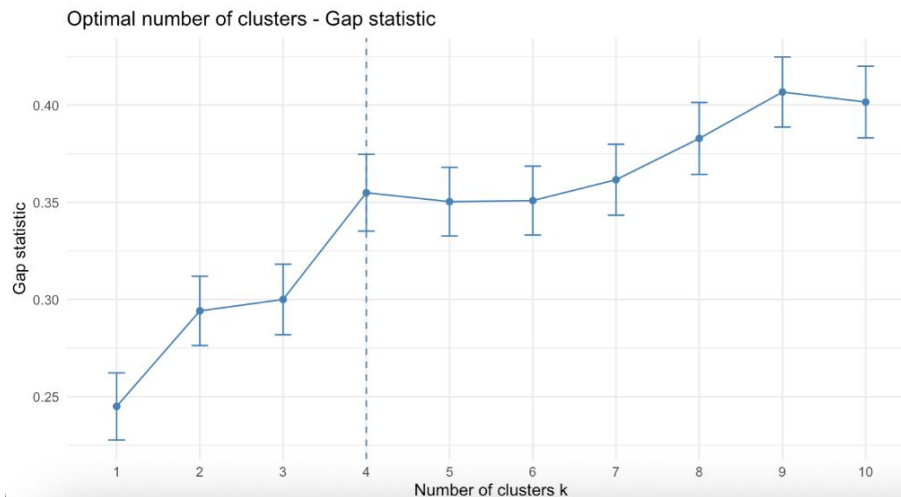
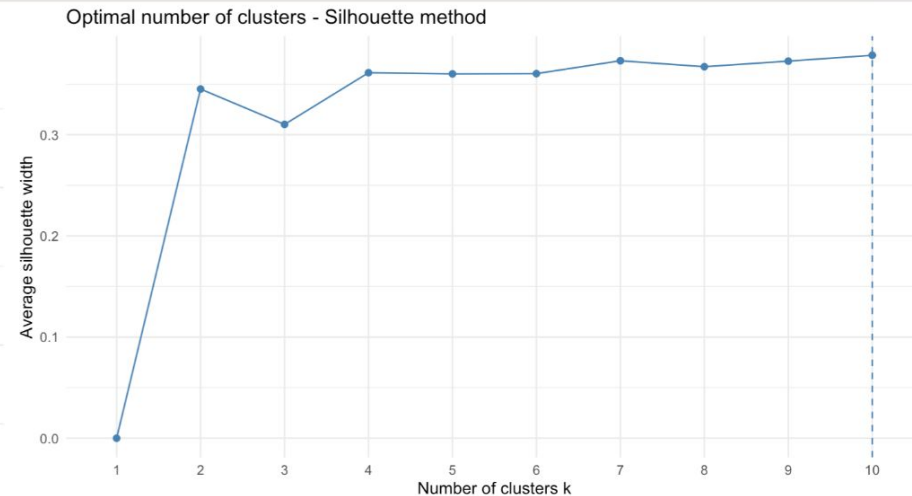
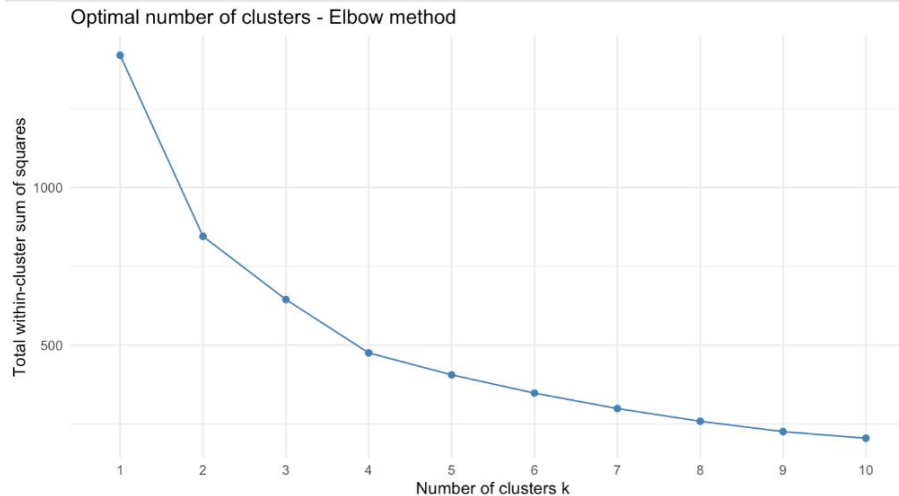


- Principal components have an interpretable tactical meaning.
- Close arrows indicate variables that tend to move together. Longer arrows indicate variables that are well represented on the PCA plane.



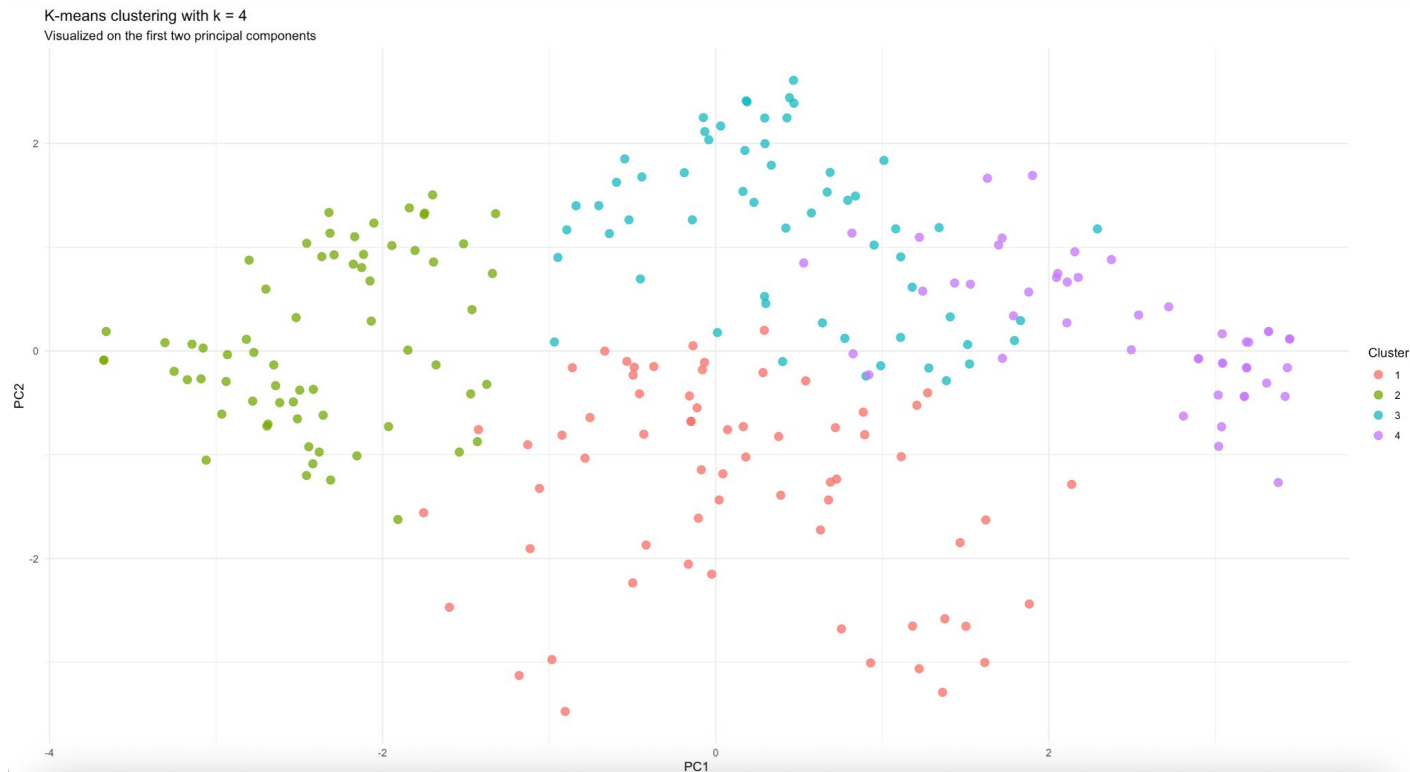
- PC1PC1 can be interpreted as overall tactical activity, because most variables load in the same direction
- PC2 captures a contrast between defensive pressure and build-up construction
- PC3 instead, adds a more specific dimension related to attacking width and chance creation

Selection of the optimal number of clusters



- The elbow method shows a strong decrease in within-cluster variability up to around $K = 4$.
- After $K = 4$, adding more clusters provides more marginal improvements.
- Silhouette analysis evaluates internal cohesion and separation between groups.
- Gap statistic compares the observed clustering structure with a random reference structure.
- $K = 4$ provides a trade-off between compactness, separation and tactical interpretability.

K-means clustering results

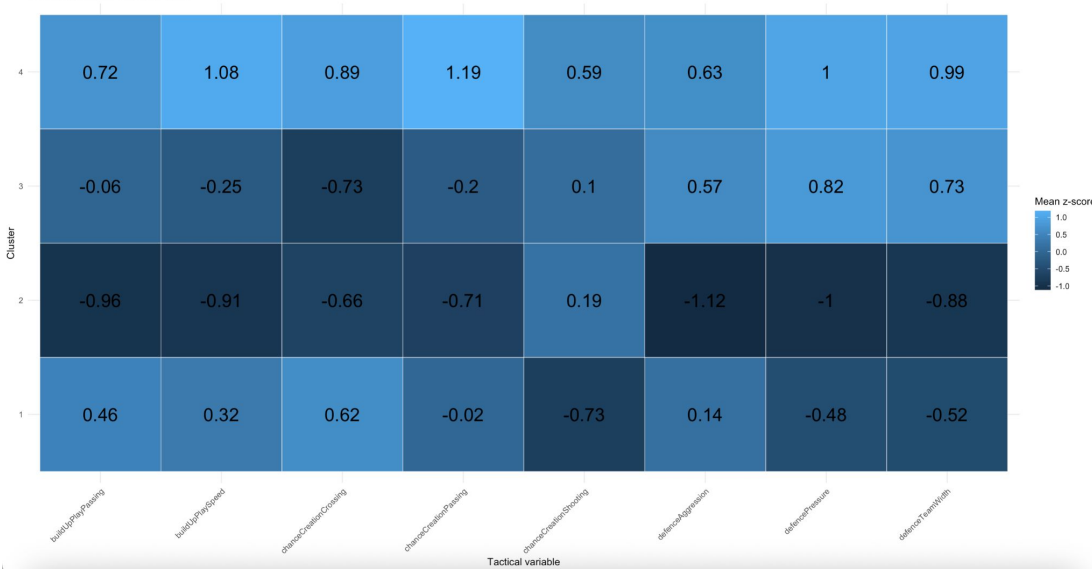


- Clustering was performed on the scores of the first three principal components. Clusters occupy different regions of the tactical space, although some overlap is present.
- Moderate separation is consistent with playing styles being continuous rather than rigid categories.

Cluster profiles and league association



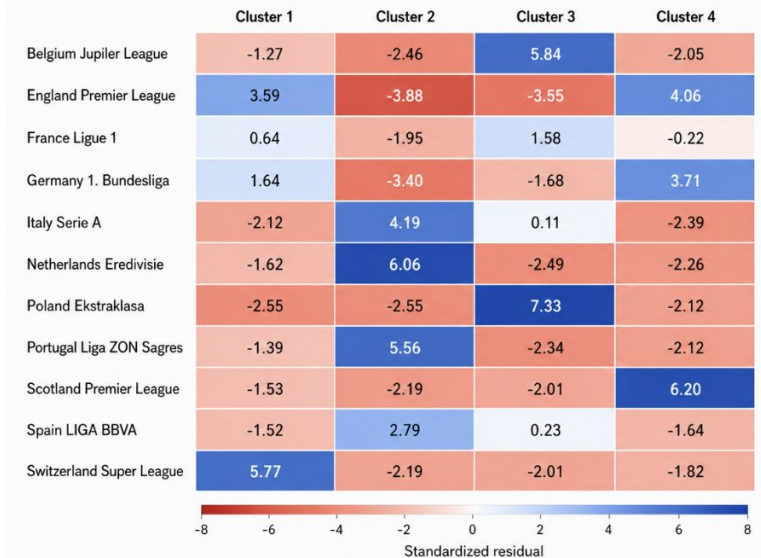
Cluster profiles based on original tactical variables
Mean standardized values by cluster



- The heatmap shows standardized average tactical variables for each cluster.
- Positive values indicate that a cluster is above the sample average for that variable.
- Negative values indicate that a cluster is below the sample average for that variable.
- Cluster 1: Possession-oriented wide build-up
- Cluster 2: Low-intensity controlled teams
- Cluster 3: Pressing and direct-transition teams
- Cluster 4: high intensity and attacking width

Standardized residuals from the Chi-square test (League × Cluster)

Positive values = over-representation; negative values = under-representation



- We applied a chi-square test of independence to assess whether the distribution of teams across tactical clusters is independent from the league.
- The chi-square test is statistically significant (simulated p-value ~ 0.0001), so league and cluster are not independent
- Values above +2 or below -2 suggest stronger deviations from independence

Research questions and dataset description



Can we spot the most significant predictors that determine the probability for a football player to become a top player?

Dataset: European Soccer Database

Focus of the analysis: Players' Attributes

Variables used: All the skills presented in the database (Physical, Technical, Tactical variables)

We conduct the following analyses separating between goalkeepers and outfield players

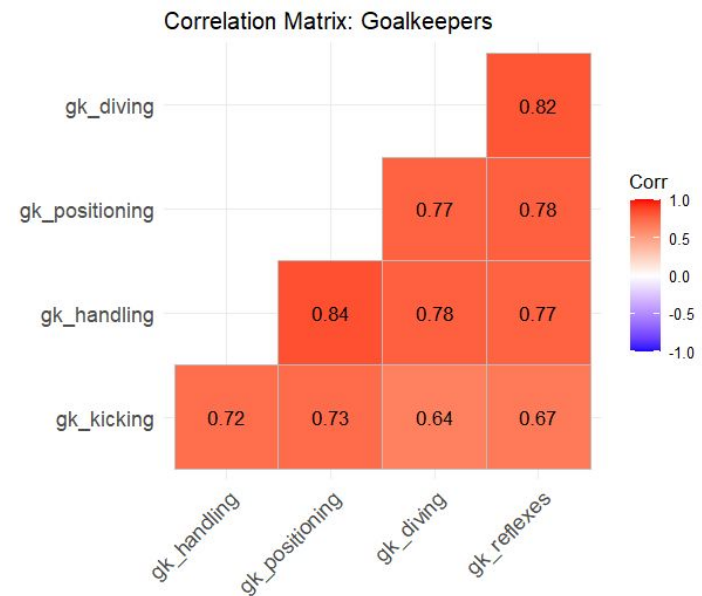
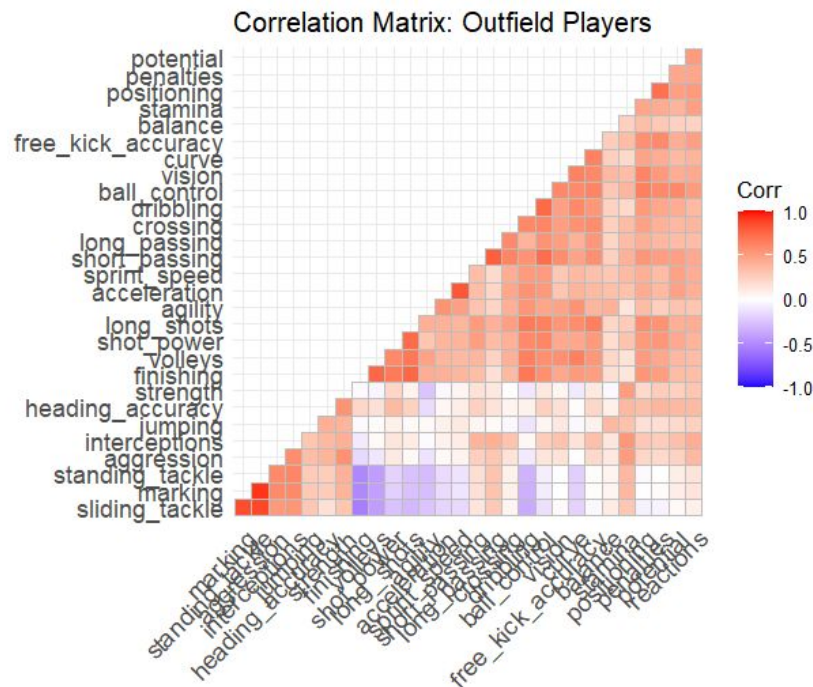


We assume that, since they perform different tasks, their probability to become a top player is driven by different factors

Top player: A players whose overall is higher than 75

Correlation matrixes

- Comparing the correlation matrixes between the two different subsets, no big differences have emerged.
- However, significant correlation emerges in both subsets between typical characteristics, preventing any OLS estimate to be reliable



Variance Inflation Factors

- When Variance Inflation Factors are considered, we find a medium level of correlation between the variables, spread between all the different sections of the two subsets.
- For outfield players, 26/28 VIF values range from 2 to 5. Only two VIF values are above the critical threshold of 5.
- For goalkeepers, VIF Values stand around 4, with the exception of *gk_kicking*.

The analysis stemming from correlation matrixes and VIF values confirms the absence of redundant variables. All the variables hold an explanatory power that cannot be neglected.



However the pervasiveness of the correlation between variables voids the OLS estimates carried out on the database. More advanced statistical methods are needed.

Ridge and Lasso regressions



Since our dependent variable is binary, logit regressions with Ridge and Lasso regularizations are performed.



To obtain 95% confidence interval, non-parametric bootstrapping (1000 iterations) is performed. The optimal λ is derived through K-fold cross-validation and is then fixed through bootstrapping iterations.

With respect to **outfield players**, Lasso selects 12 variables, which are deemed significant by Ridge too. Estimates prove to be robust also because the first 7 skills for magnitude are identical and are identically ordered in the two models.

With respect to **goalkeepers**, no variables are stable under Lasso regularization, while Ridge selects only 8 predictors. This underlines a strong multicollinearity between features. **A geometric dimension reduction is necessary to obviate this issue.**

Lasso results-Outfield palyers

Selected skills	Average coefficient	Confidence Interval	Stability
vision	+0.4008	[+0.2675; +0.5270]	YES
ball_control	+0.3530	[+0.2293; +0.4809]	YES
agility	+0.2948	[+0.1902; +0.3969]	YES
sliding_tackle	+0.2761	[+0.1661; +0.3906]	YES
volleys	+0.2469	[+0.1367; +0.3504]	YES
reactions	+0.1962	[+0.1035; +0.2899]	YES
jumping	+0.1152	[+0.0276; +0.2012]	YES
heading_accuracy	+0.1002	[+0.0181; +0.1914]	YES
aggression	-0.1097	[-0.1911; -0.0286]	YES
dribbling	-0.1340	[-0.2633; -0.0032]	YES
stamina	-0.1542	[-0.2458; -0.0604]	YES
crossing	-0.20450	[-0.3002; -0.1159]	YES

Ridge results-Outfield players

Selected skill	Average coefficient	Confidence Interval	Stability
vision	+0.3481	[+0.2623; +0.4391]	YES
ball_control	+0.2846	[+0.1921; +0.3749]	YES
agility	+0.2683	[+0.1931; +0.3459]	YES
sliding_tackle	+0.2340	[+0.1452; +0.3165]	YES
volleys	+0.2268	[+0.1420; +0.3059]	YES
reactions	+0.1862	[+0.1110; +0.2643]	YES
jumping	+0.1201	[+0.0508; +0.1908]	YES
curve	+0.1145	[+0.0388; +0.1899]	YES
heading_accuracy	+0.0948	[+0.0180; +0.1643]	YES
balance	+0.0942	[+0.0252; +0.1714]	YES
acceleration	+0.0816	[+0.0021; +0.1641]	YES
sprint_speed	+0.0811	[+0.0024; +0.1651]	YES

Ridge results-Goalkeepers

Selected skills	Average coefficient	Confidence interval	Stability
gk_diving	+0.1689	[+0.0671; +0.2674]	YES
jumping	+0.1574	[+0.0275; +0.2823]	YES
long_shots	+0.1447	[+0.0378; +0.2579]	YES
long_passing	+0.1415	[+0.0415; +0.2418]	YES
short_passing	+0.1275	[+0.0071; +0.2422]	YES
gk_handling	+0.1266	[+0.0409; +0.2182]	YES
penalties	+0.1048	[+0.0009; +0.1998]	YES
strength	-0.1203	[-0.2282; -0.0155]	YES

Sufficient Dimension Reduction



Since the estimates on our goalkeepers' database are unstable because of the high collinearity between skills, we focus on geometric dimension reduction instead of the impact of the individual components.



SDR is used also to validate our estimates on the outfield's player database. At first, we apply a Sliced Inverse Regression approach, using non-parametric bootstrapping for statistic inference (1000 iterations)

We present the results obtained for the **loadings** that define the Central Subspace. In this case the Central Subspace is constituted by a straight line since there are only two slices given by the categorical dependent variable.

To insure a valid procedure for statistic inference, we implement a **geometric alignment constraint**

SIR results-Outfield players

Selected skills	Average loading	Confidence Interval	Stability
sliding_tackle	+0.3862	[+0.2533; +0.5034]	YES
volleys	+0.3425	[+0.2365; +0.4447]	YES
vision	+0.3424	[+0.2370; +0.4456]	YES
ball_control	+0.3276	[+0.2196; +0.4358]	YES
agility	+0.2340	[+0.1442; +0.3281]	YES
reactions	+0.2083	[+0.1217; +0.2943]	YES
gk_handling	+0.1505	[+0.0408; +0.2586]	YES
curve	+0.1346	[+0.0323; +0.2387]	YES
heading_accuracy	+0.1223	[+0.0360; +0.2110]	YES
acceleration	+0.1046	[+0.0004; +0.2077]	YES
jumping	+0.0851	[+0.0055; +0.1637]	YES
balance	+0.0717	[+0.0063; +0.1428]	YES

SIR results-Goalkeepers

Selected Skill	Average loading	Confidence Interval	Stability
gk_diving	+0.3585	[-0.1611; -0.0103]	YES
long_shots	+0.2523	[-0.2423; -0.0151]	YES
gk_handling	+0.2018	[+0.0408; +0.2586]	YES
jumping	+0.1377	[+0.0055; +0.1637]	YES
acceleration	+0.1371	[+0.0004; +0.2077]	YES
ball_control	+0.0984	[+0.2196; +0.4358]	YES
heading_accuracy	+0.0901	[+0.0360; +0.2110]	YES
agility	+0.0884	[+0.1442; +0.3281]	YES
reactions	+0.0777	[+0.1217; +0.2943]	YES
curve	+0.0625	[+0.0323; +0.2387]	YES
finishing	+0.0594	[-0.2584; -0.0057]	YES
balance	+0.0417	[+0.0063; +0.1428]	YES

Sure Independence Screening



To test the relevance and the signal generated by the individual skills, we create a number of shadow variables and put the first against the noise created by the latter.

We create 10000 shadow variables for the outfield players' database, and 2000 for the goalkeepers' database in order to create an environment characterized by high dimensionality

The SIS algorithm applied to **outfield players'** skills selects 230 variables, among whom 21 true features. 18/21 of these true features reflect the ones individuated by the SIR method, including the 12 most relevant

When the algorithm is applied to **goalkeepers**, 19 variables are selected by SIS. 8/19 are true skills. In this case, there is a wider divergence when comparing with the SIR model. Only 4 of them are deemed significant also by the SIR model (gk-handling, gk_diving, jumping, agility).

Iterative Sure Independence Screening



In a context with a high multi-collinearity between variables, SIS may fail to account the impact of some skills or may overweight the importance of some marginal features.



To solve the problem of false negatives, we apply the previous algorithm iteratively following the ISIS methodology. The computational burden is obviously higher but ISIS complies with the oracle property

When ISIS is applied to outfiled players, 31 true skills are selected. This means that the algorithm deems almost every feature to be relevant in determining the probability of a non-goalkeeper to become a top-player. No fruitful considerations emerge from this result.

With respect to goalkeepers, the ISIS algorithm selects 31 variables, of whom 7 are true skills. 5/7 are identical to the ones individuated through SIS. Only 4 of them are deemed significant also by the SIR model (*gk_diving*, *long_shots*, *jumping*, *agility*).

Summary Table-Outfielders



Skill	Ridge	Lasso	SIR	SIS	ISIS
sliding_tackle	YES (0.2340)	YES (0.2761)	YES (0.3862)	YES	YES
volleys	YES (0.2268)	YES (0.2469)	YES (0.3425)	YES	YES
vision	YES (0.3481)	YES (0.4008)	YES (0.3424)	YES	YES
ball_control	YES (0.2846)	YES (0.3530)	YES (0.3276)	YES	YES
agility	YES (0.2683)	YES (0.2948)	YES (0.2340)	YES	YES
reactions	YES (0.1862)	YES (0.1962)	YES (0.2083)	YES	YES
gk_handling	YES (0.1506)	YES (0.1755)	YES (0.1505)	YES	YES
curve	YES (0.1145)	NO	YES (0.1346)	YES	YES
heading_accuracy	YES (0.0948)	YES (0.1002)	YES (0.1223)	YES	YES
acceleration	YES (0.0816)	NO	YES (0.1046)	YES	YES
short_passing	NO	NO	NO	NO	YES
interceptions	NO	NO	NO	NO	YES
jumping	YES (0.1201)	YES (0.1152)	YES (0.0851)	YES	YES

Summary Table-Goalkeepers



Skill	Ridge	Lasso	SIR	SIS	ISIS
gk_diving	YES (0.1689)	NO	YES (0.3585)	YES	YES
long_shots	YES (0.1447)	NO	YES (0.2523)	NO	NO
gk_handling	YES (0.1266)	NO	YES (0.2018)	YES	NO
short_passing	YES (0.1275)	NO	NO	NO	NO
jumping	YES (0.1574)	NO	YES (0.1377)	YES	YES
acceleration	NO	NO	YES (0.1371)	NO	NO
penalties	YES (0.1048)	NO	NO	YES	YES
long_passing	YES (0.1415)	NO	NO	YES	YES
interceptions	NO	NO	NO	YES	NO
gk_positioning	NO	NO	NO	NO	NO
gk_reflexes	NO	NO	NO	NO	NO
gk_kicking	NO	NO	NO	NO	NO

Robustness Check: Monte-Carlo simulations

To conclude the analysis on players' attributes we want to perform some robustness checks on SIR. We use Monte-Carlo simulations to test the efficiency of sufficient dimension reductions under three different assumptions. Obviously, the covariance structure is maintained.



We assume 3 different kinds of skills' distributions:

1. **The Multivariate Gaussian** distribution (serving as the baseline scenario)
2. **The Multivariate Student's t-distribution** (introducing heavy tails to model an unusually high number of both underperforming and elite players)
3. **The Log-normal distribution** (introducing severe right-skewness to model a high concentration of average values alongside a long tail of top-tier players)

To test the SIR reliability under the 3 different assumptions, we compute the alignment index. This is computed as the absolute dot product between the true coefficient vector and the first SIR eigenvector, ranging from 0 (orthogonality) to 1 (perfect alignment).

Scenario	Average alignment index	Confidence Interval (95%)
Gaussian	0,6334	[+0,4422; +0,7747]
t_Student	0,6723	[+0,4752; +0,8336]
Log_Normal	0,6401	[+0,4357; +0,8263]

Research questions and dataset description

When a bookmaker publishes a betting odd, it is implicitly stating a probability (prob = $1/\text{odds}$). The question of this analysis: are these implied probabilities well calibrated against the outcomes actually observed? And if there are systematic deviations, where are they concentrated?

Dataset: European Soccer Database

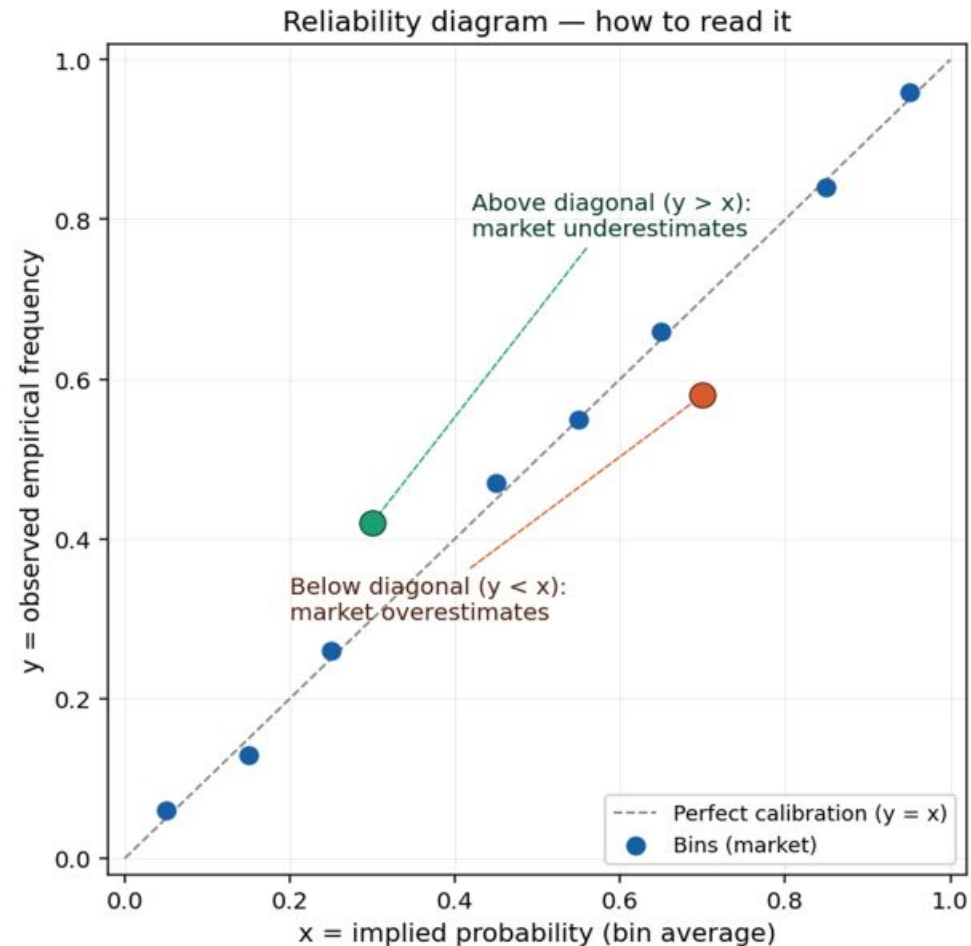
Focus of the analysis: implied probabilities vs empirical numbers

Variables used: Bookmakers quotes

The reference bookmaker is Bet365 (B365), whose average overround of 6.1% is the lowest among the six bookmakers in the dataset. Implied probabilities p_H , p_D , p_A are computed as $1/\text{odds}$ and proportionally normalized to remove the overround (so the three normalized probabilities sum to one). The outcome y is a binary indicator (1 if the event occurred, 0 otherwise).

Reliability Diagram

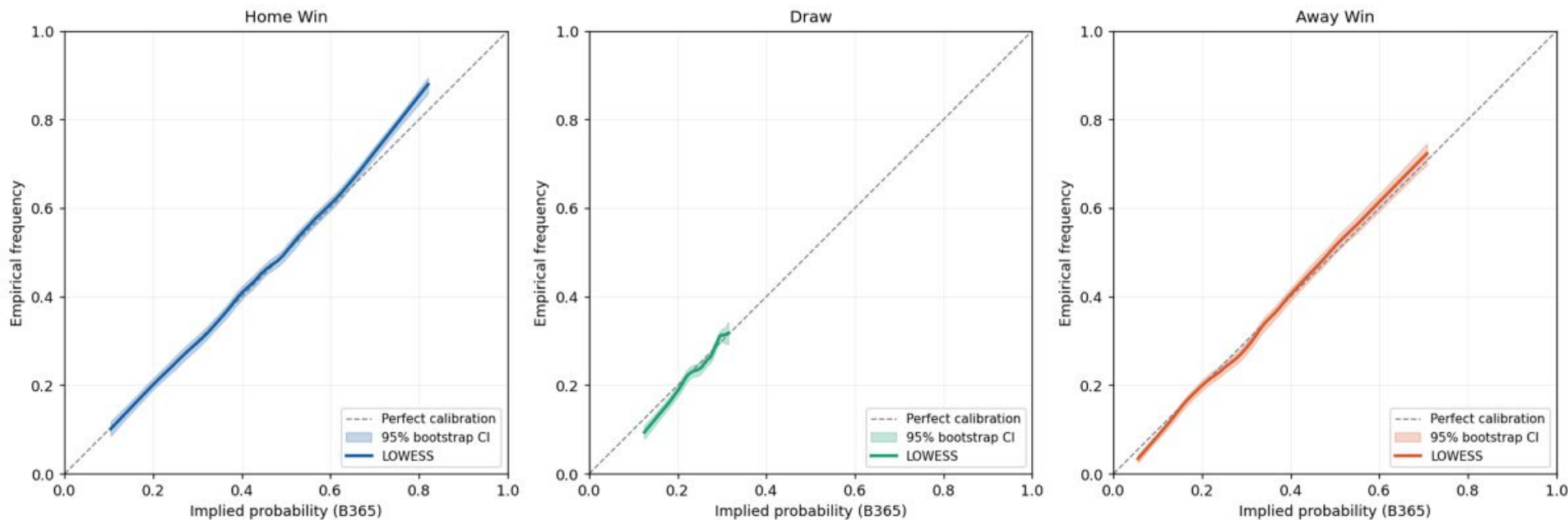
Matches are grouped into **10 bins** by implied probability. For each bin, two numbers are computed: the average implied probability (x axis) and the empirical frequency of the outcome (y axis). **A perfectly calibrated market has all points on the diagonal $y = x$.** A point above the diagonal means the market underestimates that probability; below it means the market overestimates it.



LOWESS calibration with 95% bootstrap band

LOWESS replaces the bins with a smooth curve: for each value of x , it locally estimates the empirical frequency using a weighted regression on nearby data ($\text{frac} = 0.3$). Statistical uncertainty is quantified through bootstrap (300 resamples, 95% percentile CI).

Slide 4 — LOWESS calibration curve with 95% bootstrap band (B365, $n=22,432$)



Result: Home Win and Away Win are excellently calibrated. For Draw, the LOWESS curve sits systematically below the diagonal, the market slightly overestimates the probability of a draw. The bootstrap band does not include the diagonal in some regions: **statistically significant miscalibration**.

Global calibration metrics

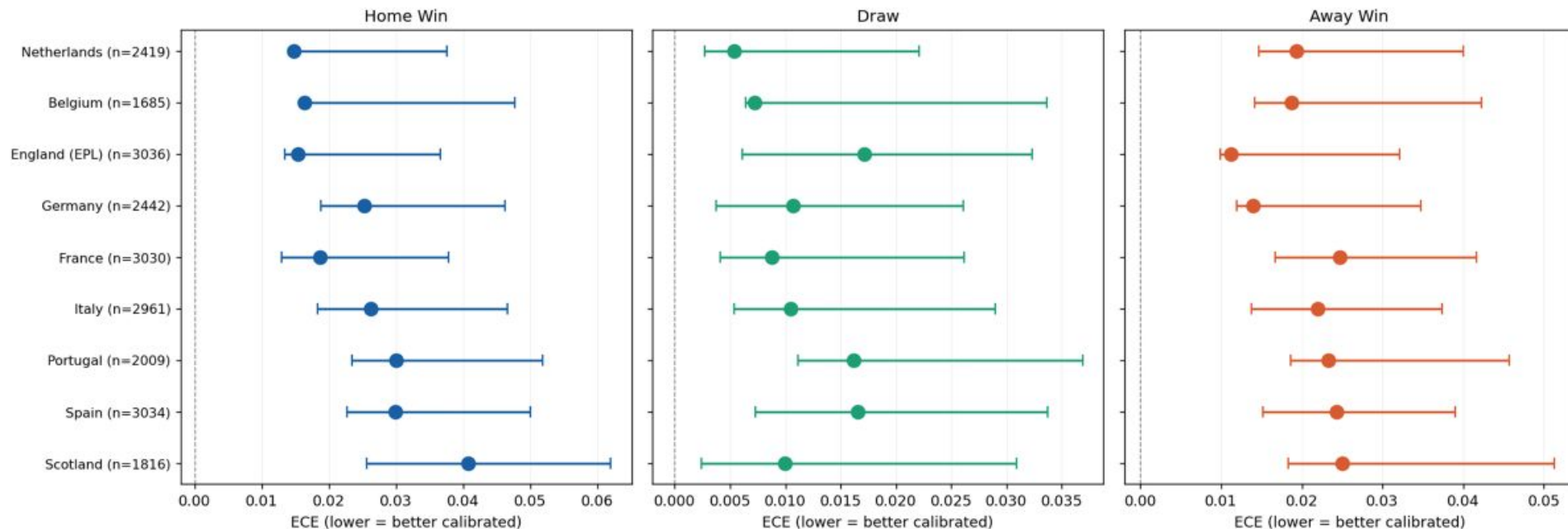
Two complementary metrics, computed on the whole dataset. **Brier score**: mean squared error between probability and outcome. **ECE** (Expected Calibration Error): weighted average of vertical distances between reliability-diagram bins and the diagonal. ECE is the most directly interpretable metric as average miscalibration in percentage points.

Outcome	Brier Score	ECE
Home win	0,2143	0,0087
Draw	0,1862	0,0054
Away win	0,1782	0,0093

ECE by league (forest plot)

For each of the 9 leagues, ECEH , ECED , ECEA are estimated together with a 95% bootstrap confidence interval (400 resamples). Leagues are sorted from best calibrated at the top to worst calibrated at the bottom (by average across the three outcomes).

Slide 5 — ECE by league with 95% bootstrap CI (400 iter.)



Main patterns: Scotland is systematically the worst calibrated; Netherlands and Belgium are surprisingly the best calibrated, even better than Spain and Italy. Confidence intervals overlap substantially, so pairwise differences are mostly not individually significant.